

# WESTERN GRAY WOLVES

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Recovery, management, genetics, and the Endangered Species Act - pre-class reading

**Purpose.** This short reading provides the background needed for the in-class gray-wolf delisting case. Focus on the distinction between biological recovery, legal listing status, and management after federal protection ends.

## Basic biology of the gray wolf

Gray wolves (*Canis lupus*) are the largest wild members of the dog family, with adults generally weighing about 40–175 lb (18–80 kg) depending on sex and geographic region. They are highly social and usually live in packs organized around a breeding pair and their offspring; packs in the western United States are often relatively small, although they can occasionally contain 20 or more animals. Wolves are primarily predators of medium- and large-bodied mammals such as elk, deer, moose, and other ungulates, but they are opportunistic and can use a wide variety of food resources. [1]

Wolves are also unusually flexible in both habitat use and social behavior. They can occupy forests, grasslands, shrublands, mountains, and other landscapes as long as adequate prey is available and human-caused mortality is not too high. Young wolves frequently disperse from their natal packs, sometimes traveling hundreds of miles before establishing a territory or joining another pack. This capacity for long-distance dispersal allows populations to recolonize vacant habitat and maintain demographic and genetic connections across large landscapes. Their high reproductive potential, flexible pack structure, and ability to disperse are major reasons wolf populations can recover relatively quickly when mortality is reduced. [1]

## A species nearly removed from the West

Gray wolves once occupied most of North America, including much of the western United States. European settlement was followed by widespread poisoning, trapping, shooting, and government-sponsored predator-control programs. By the 1930s, wolves had been essentially eliminated from the western United States. Federal protection began before passage of the modern ESA; the Northern Rocky Mountain wolf was listed in 1973, and in 1978 FWS consolidated earlier listings into broader gray-wolf protection across most of the lower 48. [1,3]

## Wolves return

Recovery occurred through a combination of natural recolonization from Canada and deliberate reintroduction. In 1995-1996, Canadian wolves were released in Yellowstone National Park and central Idaho. Those populations grew, dispersed, and established across the Northern Rockies. Wolves subsequently expanded into Oregon, Washington, California, and Colorado. By the end of 2022, FWS estimated about 2,797 wolves across the western United States. [1]

## What does “recovered” mean?

The Northern Rocky Mountain recovery program ultimately used a benchmark of at least 300 wolves and 30 breeding pairs distributed across Idaho, Montana, and Wyoming for three consecutive years, together with genetic exchange among recovery populations. Those numerical benchmarks were exceeded. But recovery-plan criteria and ESA listing status are not identical: FWS must still evaluate whether the relevant entity meets the legal definitions of endangered or threatened based on current threats and the best available scientific and commercial data. [1,3]

## Recovery changes the geography of the population

The NRM DPS once described a relatively isolated recovered population. Recovery itself changed that geography. As wolves expanded into Oregon, Washington, and California, the gap between the NRM and western wolves disappeared. In its 2024 assessment, FWS concluded that the old NRM population no longer satisfies the DPS discreteness requirement. FWS nevertheless found that a broader western U.S. population could qualify as a valid DPS because it is distinct from other major wolf groups and because losing it would create a major gap in the species’ North American range. [2]

## Why connectivity and genetics matter

Wolves are highly mobile, and successful dispersers can link distant packs. Connectivity can recolonize empty habitat, rescue small populations demographically, and move genetic variation among populations. Census size is not the same as effective population size ( $N_e$ ): in a social species such as wolves, relatively few animals may contribute offspring each generation. vonHoldt et al. found that wolf  $N_e$  estimates averaged only about 5.2-9.3% of census size and documented declining heterozygosity through recent decades in the Northern Rockies. [4] FWS, meanwhile, concluded that contemporary western wolves remain well connected and genetically diverse, with very low modeled risk of short-term inbreeding problems. [1]

## The major contemporary threat is people

FWS identifies human-caused mortality as the dominant current stressor on western wolves. Sources include regulated hunting and trapping, lethal control after livestock depredation, and illegal killing. Disease can produce localized short-term declines, while habitat and prey remain broadly available. [1] This makes regulation central to the recovery question: western wolves are biologically capable of rapid growth and recolonization, but their future abundance is strongly influenced by human management.

## What changes when federal protection ends?

Delisting does not mean that management ends. It means that federal ESA protections no longer provide the primary framework and management shifts principally to states and Tribes. State objectives and rules can differ substantially. Idaho and Montana adopted more permissive harvest regulations beginning in 2021; those changes helped trigger petitions asking FWS to restore federal protection. FWS ultimately concluded in 2024 that a Western U.S. DPS did not warrant listing as endangered or threatened. [1,2,6]

## Why legal status and biological population boundaries do not match neatly

The regulatory history of western wolves is unusually complicated. Congress intervened in 2011 to direct reinstatement of Northern Rockies delisting, and Wyoming was ultimately delisted in 2017. A broader lower-48 delisting took effect in 2021 but was vacated by a federal court in 2022. The result is a legal map that does not perfectly match the biological population structure described by FWS. Current FWS guidance states that wolves outside the Northern Rocky Mountains remain federally protected following the 2022 court order, while NRM wolves remain outside ESA protection. [3,7]

## Why the wolf case is so useful for studying the ESA

Between 2007 and 2017, the ESA status of at least some gray-wolf population segment changed 16 times. Rules were issued, challenged, vacated, reinstated, and in one major instance Congress intervened directly. Martin (2020) interprets this history through the lens of adversarial legalism; the nonpartisan CRS report presents the underlying sequence of agency rules, court decisions, and statutory interpretation issues. [3,5]

## Sources

[1] U.S. Fish and Wildlife Service. 2023. Species Status Assessment for the Gray Wolf (*Canis lupus*) in the Western United States, Version 1.2. Especially Executive Summary and Chapters 2-4.

[2] U.S. Fish and Wildlife Service. 2024. Species Assessment and Listing Priority Assignment Form / 12-month finding for gray wolf in the Northern Rocky Mountains and Western United States (information current Jan. 4, 2024).

[3] Congressional Research Service. 2020. The Gray Wolf Under the Endangered Species Act (ESA): A Case Study in Listing and Delisting Challenges. R46184.

[4] vonHoldt, B.M. et al. 2024. Demographic history shapes North American gray wolf genomic diversity and informs species' conservation. *Molecular Ecology* 33:e17231. doi:10.1111/mec.17231.

[5] Martin, D.R.P. 2020. Prolonged gray wolf endangered species act listing fluctuation and policymaker response in the United States. *Biological Conservation* 242:108408.

[6] U.S. Fish and Wildlife Service. 2024. 'Service Announces Gray Wolf Finding and National Recovery Plan.' Feb. 2, 2024.

[7] U.S. Fish and Wildlife Service. Gray Wolf Recovery News and Updates (current legal-status summary; accessed Aug. 2026).